COMPUTER ENGINEERING

QUEZON CITY

COURSE CODE/TITLE:	CPE-010 - Data Structures and Algorithms	**SECTION:**	CPE21S2
DATE PERFORMED:	8/26/26	**DATE SUBMITTED:**	8/26/26
NAME:	SUBALISID, JAPPETH ROYCE B.	**INSTRUCTOR:**	ENGR. JIMLORD M. QUEJADO

ACTIVITY NO. ___ [ACTIVITY NAME]

1. PROCEDURE OUTPUT

Problem Definition

A sequence is called symmetric if the sequence of values is the same when read from the beginning to the end and from the end to the beginning.
For example:

1 2 3 2 1 is symmetric.
1 2 3 4 1 is not symmetric.
5 5 is symmetric.

Given a sequence of positive integers, the task is to determine whether the sequence is symmetric or not.
You are required to use data structures to store and process the values in the sequence.
Your task is to use two data structures for your solution:

Stack
Queue

However, the two data structures must have different internal representations, either:

Array-based
Linked List

Example: If your stack is array-based, your queue must be implemented using a linked list, and vice versa.

The sequence must be inserted into both the Stack and Queue. You must then use the behavior of the two data structures to determine whether the sequence is symmetric.

The user is prompted to enter the values of a sequence. Your program must then determine and display whether the given sequence is symmetric.

Sample Test Data

Input Sequence	Symmetric?	Output (Console)	Remark
1 2 3 2 1	Yes		
1 2 3 4 5	No		
1 2 2 1	Yes		
7	Yes		
1 2 3 3 2 1	Yes		
1 2 3 4 3 2 5	No		

Expected Approach

The solution should take advantage of the different behaviors of the two data structures:

The Stack returns values in Last-In, First-Out (LIFO) order.
The Queue returns values in First-In, First-Out (FIFO) order.

For every value entered by the user, the value should be stored in both data structures.

The program should then retrieve values from the Stack and Queue and compare them.

If every pair of values is equal, the sequence is symmetric. If at least one pair differs, the sequence is not symmetric.

Submission Requirements:

1. Activity template containing the screenshots and analysis of the function you created to solve the problem. (Code snippet only, not the whole code)

2. Code outputs inside the sample test data.

3. .cpp and .h files

STACK IMPLEMENTATION:

1. Push values into stack
2. Read stack backward, save into other stack
3. Read stack forward, save into string
4. Compare string, return bool

- Since we're using an array to read the stack, we can take advantage of a for loop and the Top variable to read the array forwards and backwards using the header file. We can then store these arrays into string and compare the equality of the two, if it returns true, it means that it is symmetrical.

```
bool compare(){ //compareing forward and backward
    if(isEmpty()) {
        std::cout << "\nStack is empty!!!";
        return false;
    }
    std::string forward, backward; //storing array into strings
    for(int i = top; i != -1; i--) { //backward read
        backward += stackArr[i];
    }
    for(int i = 0; i <= top; i++) { //forward read
        forward += stackArr[i];
    }
    return forward == backward; //compare strings
}
```

Input Sequence	Symmetric?	Output (Console)	Remark
1 2 3 2 1	Yes	Stack: [1 2 3 2 1] The stack is symmetrical.	Success
1 2 3 4 5	No	Stack: [5 4 3 2 1] The stack is not symmetrical.	Success
1 2 2 1	Yes	Stack: [1 2 2 1] The stack is symmetrical.	Success
7	Yes	Stack: [7] The stack is symmetrical.	Success
1 2 3 3 2 1	Yes	Stack: [1 2 3 3 2 1] The stack is symmetrical.	Success
1 2 3 4 3 2 5	No	Stack: [5 2 3 4 3 2 1] The stack is not symmetrical.	Success

QUEUES

1. Enqueue the data into a queue
2. Compare frontptr to backptr
3. Dequeue front
4. Compare new front to the second to the last
5. Repeat steps until remaining node is 0 or 1
6. If it fails to equalize until the end, return false

-I can't seem to process how to use the FIFO behavior of the queue to get the reverse order of the queue, so I thought maybe I could utilize a doubly linked list to get values from the back of the queue to be able to compare the front and the back.That way we can just dequeue until we get a false '==' result to show that the queue is not symmetrical.

Input Sequence	Symmetric?	Output (Console)	Remark
1 2 3 2 1	Yes		
1 2 3 4 5	No		
1 2 2 1	Yes		
7	Yes		
1 2 3 3 2 1	Yes		
1 2 3 4 3 2 5	No		

2. ANSWERS TO SUPPLEMENTAL ACTIVITY

Copy and paste the questions from the lab and include your answers after.

3. ARTIFICIAL INTELLIGENCE (AI) USE DISCLOSURE STATEMENT

I hereby declare that artificial intelligence (AI) tools were used, if applicable, in the preparation of this academic work. The following indicates the nature and extent of AI assistance:

Tools Used: _____

Examples: ChatGPT, Microsoft Copilot, Grammarly, QuillBot, etc.

Purpose of Use: _____

Examples: grammar correction, idea generation, formatting, summarization, code assistance, explanation of concepts, etc.

Extent of Use: _____

Explain how AI was used and confirm that it did not replace original thinking, analysis, authorship, or completion of the required work.

COMPUTER ENGINEERING

QUEZON CITY

By submitting this activity, I affirm that my use of AI complies with T.I.P.'s AI usage policy and does not exceed the permitted limits for similarity, automated content generation, or academic assistance.

I understand that any false, incomplete, or misleading disclosure may be subject to investigation in accordance with due process and may result in sanctions for violation of existing T.I.P. policies.

Lab activities, templates, and related instructional materials shall not be reuploaded, reproduced, distributed, or shared with external organizations or third parties without the prior express written permission of the Technological Institute of the Philippines.